Fourth Eon Biosecurity

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Building adaptive biosecurity infrastructure that evolves with AI and bioengineering

Summary

Life on Earth is defended by a dangerous assumption: that we can safeguard biology with a historical archive of known threats. AI is breaking that assumption. For the first time, novel biological agents can be engineered faster than defenses can catalog them – a biological zero-day. We are one synthesis away from a devastating engineered pandemic.

What's needed is an adaptive approach to biosecurity that can evolve to keep pace with advances in AI-enabled bioengineering. Fourth Eon Biosecurity ("Fourth Eon") will be a 501(c)(3) nonprofit and serve as the permanent home for developing AI-resilient biosecurity infrastructure. Our first R&D program, SCREEN, introduces adaptive function-based screening: an AI agent orchestrating specialized biological models to evaluate whether a sequence encodes a harmful function. We have secured a $2M anchor commitment from Sentinel Bio, and are seeking an additional $3M to complete the 18-month Tractability Phase. The screening standards set over the next two years will define the global biosecurity baseline for a generation. We are building the screening infrastructure those standards will require.

The Challenge

Problem

Powerful general-purpose AI models are lowering the barrier to engineering harmful biology while specialist tools raise the ceiling of what is achievable.[1,2] In controlled red-teaming, researchers used three protein models to redesign 72 proteins of concern into over 76,000 synthetic variants. Individual tools missed 30-77% of variants due to how far they had diverged from known sequences. After patching the tools, 3% of the highest-confidence designs remained undetected.[3] A March 2026 follow-up found that fragments of AI-redesigned variants posed further detection challenges.[4] Current screening relies on sequence-based identification of known hazards and cannot reliably detect novel AI-generated threats.

Why Now

Three forces are converging. Capability is rising as AI biodesign tools improve and synthesis costs fall. The skill barrier is eroding as LLMs and AI agents lower the operator threshold, democratizing access to dual-use bioengineering. And policy is shifting to statutory mandates: US EO 14292 directed OSTP to revise oversight policy; the Biosecurity Modernization and Innovation Act (S.3741) would establish mandatory screening with civil penalties;[5] and the EU's proposed Biotech Act would require built-in screening for benchtop devices.[6] This environment requires robust, AI-resilient biosecurity infrastructure that does not yet exist.

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[1] Watson JL, et al. *Nature.* 2023;620(7976):1089-1100.

[2] Baker D, Church G. *Science.* 2024;383(6681):349.

[3] Wittmann BJ, et al. *Science.* 2025;390(6768):82-87.

[4] Wittmann BJ, Wheeler NE, et al. *bioRxiv.* 2026. doi:10.64898/2026.03.04.709671

[5] Biosecurity Modernization and Innovation Act, S.3741, 119th Cong(2026).

[6] European Biotech Act. COM(2025) 1022 final. Dec 16, 2025.

The Opportunity

Thesis

Biosecurity infrastructure is both a structural market failure and a timing problem. DNA synthesis providers bear the cost of screening, and implementing stronger safeguards erodes their competitive position. Academia cannot deploy industrial-grade software at scale. And government procurement cannot keep pace with rapidly advancing AI biodesign. Once legislation mandates robust, AI-resilient screening, providers will need compliant tools. Fourth Eon's goal is to have validated screening infrastructure ready when mandates hit; philanthropy bridges this gap.

Biosecurity developers are exploring various approaches to improve AI-resilience; however, no one is currently building an adaptive, AI-native screening system from the ground up that dynamically incorporates emerging biological AI capabilities. Fourth Eon's modular screening architecture allows new models to plug in as they are released. The organizational form is modeled on ISRG (27 employees, ~$9.6M revenue, 700M+ websites with Let's Encrypt) and MITRE (CVE database and ATT&CK framework), combining lean infrastructure-first execution with honest-broker positioning. The 501(c)(3) will be mission-locked by charter, and cannot be redirected or subsumed.

R&D Program

Fourth Eon builds on advancing AI capabilities to accelerate development of AI-resilient access controls. Four modular tracks branch from one core R&D program focused on adaptive, function-based biological threat detection. Phase 1 (Tractability) concentrates on SCREEN. If it fails, we stop and document why. If it succeeds, subsequent programs advance as SCREEN achieves predefined milestones.

CORE R&D: Adaptive, function-based biological threat detection – methods, standards, and test sets			
SCREEN	**SAFEGUARD**	**SECURE**	**SHIELD**
Agentic function-based DNA synthesis screening. Live pilot underway.	Safety constraints for AI biodesign tools and frontier language models.	Firmware-level security for benchtop DNA synthesizers.	Protocol-level biohazard detection for cloud labs.
Now	**Next**	**Horizon**	**Horizon**

SCREEN employs an agentic analysis pipeline that invokes specialized biological models (structure, function, binding, toxicity, etc.) to evaluate whether a sequence encodes a harmful function. The agentic design is a critical decision: the system adapts to the inputs, dramatically reducing compute costs. Modularity means new models plug in without redesigning the pipeline. The capabilities developed through this core R&D also support SAFEGUARD, SECURE, and SHIELD. See Appendices A & B.

Traction

- **SCREEN prototype achieved 100% classification accuracy** on the 999-sequence NIST screening benchmark in internal testing, exceeding the 99.8% best individual tool performance[7]
- **Live pilot underway** with a commercial DNA synthesis provider on real-world order data. An 8,000-sequence run matched 100% of provider flags while identifying dozens of hazardous sequences that existing tools missed
- **U.S. provisional patent application** filed 2025, with claims to core adaptive screening methods
- **Perspective paper published** with expert consensus on the need for function-based screening[8]
- **Early-stage support** from Sentinel Bio, OpenAI, and the Chan Zuckerberg Biohub
- **$2M anchor grant** secured from Sentinel Bio; raising a further $3M to complete Phase 1

[7] Laird TS, et al. *Appl Biosaf*. 2025. doi: 10.1177/15356760251401228
[8] Abel G, et al. *Front in Bioeng and Biotechnol.* 2026;14. doi: 10.3389/fbioe.2026.1832724.

Key Stakeholders

Fourth Eon will operate as a neutral infrastructure provider. Key stakeholders include defense and intelligence agencies (FBI, DARPA, IARPA), biotech companies (DNA synthesis providers, AI model developers, benchtop device manufacturers, cloud labs), regulators and standards bodies (NIST, ISO, ASPR, OSTP), NGOs and multilaterals (IBBIS, IGSC, SBRC, EBRC, WHO), and screening developers.

Budget

Phase	Capital	Timeline
Phase 1 - Tractability	$5M	18 months
Phase 2 - Scale	$10-20M	24-36 months
Phase 3 - Fully Operational	$35-50M cumulative	4.5-6.5 years

Phase 1 is supported by 100% philanthropic funding. Revenue emerges in Phase 2 (~$500K-$2M/year) and scales to $5-$15M/year at maturity. See Appendix C.

Team & Roles

Fourth Eon's core team spans the full stack, from molecular biophysics, biosecurity, and AI safety to institutional design and deep tech commercialization.

Chris Ganje - Executive Director. 20 years bridging advanced technology and organizational adoption. Defense-related collaborations with Georgetown's CSET, Harvard's Belfer Center; Innovate UK Council Member; Cambridge Centre for Science & Policy; DARPA Switch Policy Working Group Advisor.

Gary Abel, PhD - Chief Scientist. 16 years biotech R&D; commercial DNA sequencing instrument launch (startup acquired by PacBio). Contributing Scholar, Johns Hopkins Center for Health Security. Lead author on the consensus paper *Beyond Sequence Similarity*.[8] Co-developed SCREEN prototype.

James Black, MD, PhD - Founding Scientific Advisor. Biosecurity, evolutionary biology, AI safety testing across frontier LLMs and biodesign models.[9] Co-developed SCREEN prototype. Contributing Scholar, Johns Hopkins CHS. Accepted position at frontier AI lab; remains actively contributing.

Impact & Outcomes

Short-Term (18 months): 501(c)(3) operational with Governance Board and Scientific Advisory Board; developed and validated screening platform on real provider data and challenging test sets; peer-reviewed publications on screening standards and methods; completed provider pilot. Decision point: if the architecture works, deploy and scale up; if not, stop and document why.

Long-Term (Years 2-5): SCREEN deployed at provider scale; SAFEGUARD, SECURE, SHIELD prototypes under development; ongoing international standards contributions; approaching self-sustaining operations through provider contracts and certification. Findings published on an ongoing basis, consistent with responsible disclosure.

The Ask

We are seeking $3M over 18 months, alongside Sentinel Bio's $2M commitment, to establish the institution, assemble the R&D team, and complete the Tractability Phase.

[9] Black JRM, et al. arXiv:2511.19299. 2025.

Appendix A: Organization, R&D Program, & Responsible Research

Fourth Eon Biosecurity will be established as an independent 501(c)(3) not-for-profit applied R&D organization, with a fully-controlled commercial subsidiary for holding IP that may be involved in future commercialization. The structure is mission-locked by charter: it cannot be acquired, redirected, or otherwise subsumed, and revenue from any future commercialization will be reinvested to support continued R&D and deployment. The Governance Board provides oversight, with expert guidance from the Scientific Advisory Board (5-10 members).

The development of adaptive function-based screening is coupled to foundational and applied research efforts that advance the AI biosecurity field:

- **An ontological framework** for precisely defining and categorizing functions of concern
- **Construction of challenging datasets** spanning natural, engineered, and AI-designed sequence spaces, for testing models and screening methods
- **Training classifiers** that discriminate between functional and nonfunctional sequences and detect specific types of hazardous functions
- **Biological foundation model interpretability** for molecular property prediction and function-based screening
- **Agentic language-model fine-tuning and specification** for use in adaptive screening workflows
- **Wet-lab characterization** of safe AI-generated proxy proteins, building on the NIST TEVV methodology[10] and Align Bio's GROQ-SEQ platform[11]

Fourth Eon's R&D program is designed to convert AI capability gains directly into advances in biosecurity. The core technical work of function-based screening falls squarely in the domains where current frontier models already provide significant researcher uplift. Our organizational strategy incorporates this in three ways: (1) deliberately automate R&D tasks where models are capable, reserving human bandwidth for high-leverage activities; (2) prioritize measurement infrastructure (test sets, evals, benchmarks, performance metrics) that multiply the value of AI labor applied to the problem; and (3) treat inference compute as a budget line that grows as a fraction of total spend over the grant period. The aim is to enable a small, agile, high-caliber team to operate at the speed and scale the biosecurity threat landscape now demands.

Fourth Eon Biosecurity is committed to a **Safe and Responsible Research Framework**, including regular biosecurity reviews with a risk assessment and mitigation strategy, strict information security measures, responsible disclosure of sensitive findings, engagement with expert stakeholders, and use of benign proxy molecules for wet-lab experiments. The research is carried out with oversight from the Governance Board, with authority to restrict or delay publication of sensitive findings.

[10] Ikonomova, S. P., et al. bioRxiv. 2025. doi:10.1101/2025.05.15.654077

[11] Cortade, D., et al. Zenodo. 2024. doi:10.5281/zenodo.13909104

Appendix B: Focus Areas

SCREEN (Now): SCREEN develops a novel sequence screening architecture: an agentic analysis pipeline that autonomously invokes specialized biological models and tools, including protein structure prediction, functional annotation, binding affinity estimation, and toxicity assessment, to evaluate whether a submitted sequence encodes a harmful biological function. Unlike sequence-matching approaches, this method can assess threats that bear no homology to any known pathogen or toxin. The agentic design is not a marketing choice but a critical design decision: the ability to selectively invoke tools in an iterative analysis process dramatically reduces compute requirements, saving the most intensive tools for when they are needed. New biological models can be incorporated as they emerge without redesigning the pipeline. Initial deployment targets DNA synthesis providers, with expansion paths to peptide and protein synthesis providers and selected CRO workflows.

SAFEGUARD (Next): SAFEGUARD adapts the function-based approach of SCREEN to develop robust safety filters for AI biodesign tools and frontier model biosafety layers, preventing unauthorized generation of harmful designs upstream. SAFEGUARD begins active development only after SCREEN has validated the core detection architecture. The same function-prediction capabilities that power SCREEN can be repurposed for design safety: if you can detect whether a protein is harmful after it has been designed, you can embed that detection upstream. This is the bridge from SCREEN to SAFEGUARD.

SECURE (Horizon): SECURE embeds function-based screening directly into the control software and firmware of benchtop synthesizers. With gene-length benchtop capability forecast by approximately 2028-2030 and only 54 units (5 kbp capability) to 92 units (1 kbp capability) projected worldwide by 2030,[12] the integration challenge is manageable if we start now. The proposed European Biotech Act would require built-in screening mechanisms for devices sold in Europe. AI-resilient screening must be incorporated into distributed synthesis hardware.

SHIELD (Horizon): SHIELD extends function-based screening to cloud laboratories through protocol-level analysis. Cloud labs and contract research organizations (CROs) are creating fully automated biology capabilities with minimal human oversight, creating a new frontier of biosecurity exposure and a new level of complexity that current screening infrastructure does not address. Screening cloud lab orders involves predicting whether a protocol is likely to produce a biohazard – a task well-suited to adaptive agentic analysis methods.

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[12] Langenkamp M. *Securing Benchtop DNA Synthesizers.* Institute for Progress; 2024.

Appendix C: Budget

Phase 1 - Tractability (18 months):

Category	Spend
Personnel - 6-8 FTEs (~55%)	$2.75M
Compute, AI, infrastructure (~20%)	$1.00M
Experimental data collection (~5%)	$0.25M
Integration, validation, pilot (~4%)	$0.20M
Ecosystem engagement (~3%)	$0.15M
Business, legal, & overhead (~8%)	$0.40M
Contingency (~5%)	$0.25M
Total	**$5.0M**

Phase 2 - Scale (24-36 months): $10-20M. SCREEN scaling; SAFEGUARD development; SECURE/SHIELD scoping. 12-20 FTEs. Revenue: $500K-$2M/year.

Phase 3 - Fully Operational (3-5 additional years): $35-50M cumulative. All four programs. Flywheel: $5-$15M/year from provider contracts, certification, government contracts. Initial capital: philanthropic grants, government grants, possible future equity financing through a fully controlled commercial subsidiary.